MIL-T-55603A(EL)
31 October 1975
superseding
MIL-T-55603(EL)
5 March 1971

MILITARY SPECIFICATION
TEST SET, TELEPHONE AN/PTM-7( )

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers Test Set, Telephone AN/PTM-7( ) which is portable, self powered, and provides the facility for the field maintenance of the special purpose cable assembly CX-11230( )/G and Restorer Pulse Form TD-206( )/G connected in Pulse Code Modulation (PCM) - Time Division Multiplexer Cable Transmission Systems. Test Set, Telephone AN/PTM-7( ) consists of the following: (See 6.1 and 6.5)

Test Set, Telephone TS-1323( )/PTM-7 with combination case.
Headset-Microphone H-91( )/U
Lead, Test CX-10354( )/PTM-7
Lead, Test CX-10355( )/PTM-7
Simplified Instruction Card
Sling, Carrying

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of Invitation For Bids or Request for Proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box Fiberboard, Corrugated, Triple Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Contd)

PPP-P-291	Paperboard, Wrapping and Cushioning
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-S-1698	Slings, Carrying, Bag and Case
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (Power, Control, and Audio Frequency), General Specification For
MIL-P-3938	Pallet, Material Handling, Hardwood, Stringer Construction, 4-Way (Partial)
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-H-12576	Headset-Microphones H-91( )/U and H-92( )/U and Handset-Headset H-144( )/U
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-15011	Pallet, Material Handling, Wood, Post Construction, 4-Way Entry
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible For Packaging
MIL-I-46058	Insulating Compound, Electrical (for Coating Printed Wiring Assemblies)
MIL-C-55583	Cable Assembly Special Purpose Electrical CX-11230( )/G and Cable Assembly Adaptor CX-10734( )/G

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40 in x 48 in 4-Way Partial and 4-Way Pallets)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

STANDARDS (Contd)

MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Enviornmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DL-SC-A-339551	Cable Assembly Adaptor CX-10734/G
SM-C-528098	Simplified Instruction Cards
SC-DL-525135	Cable Assembly, Power Electrical CX-11727/G
SC-DL-528168	Test Set, Telephone AN/PTM-7( )
SC-DL-531012	Cable Assembly, Radio Frequency CG-1040( )/U
SC-DL-531016	Cable Assembly, Radio Frequency CG-2437/TCC(ft)
SC-DL-531017	Cable Assembly, Radio Frequency, CG-2438/TCC (ft in)

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM-11-6625-648-12	Operator and Organizational Maintenance Manual - Test Set, Telephone AN/PTM-7

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First Article. When specified by the contract or purchase order, the contractor shall furnish First Article Units in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data List SC-DL-528168 including the requirements for parts, materials, and processes thereon.

3.2.1 Headset, Microphone H-91( )/U. The Headset-Microphone H-91( )/U shall comply with MIL-H-12576 (see 4.4).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4)

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification (see 4.4).

3.5.2 Visibility. Wherever practicable, parts and components shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Test Set, Telephone TS-1323( )/PTM-7 shall be serial numbered.

3.6 Electrical Characteristics.

3.6.1 Talk-Listen capability (See 4.8.5). A cable system talk and listen facility using Headset-Microphone H-91( )/U with Test Set, Telephone TS-1323( )/PTM-7 shall provide intelligible voice communication.

3.6.2 Receive amplifier (See 4.8.10). The hybrid-receiver amplifier combination shall maintain 40 to 270 millivolts RMS output at 1KHZ across 275 ohms for input signal levels between 300 millivolt and 16 volt RMS.

3.6.3 Transmit amplifier (See 4.8.11). A 100 millivolt RMS, 1KHZ signal applied to the 50 ohm input of the Transmit Amplifier (E64 to E52) shall produce a 1KHZ signal between J23 and ground with a minimum amplitude of three volts RMS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.6.4 Ringing (Signal) detector (See 4.8.12). An input of 300 mv to 16 volts RMS originating from an oscillator whose frequency is 1600 HZ ± 2% shall cause the signal detector to light the call lamp and sound the buzzer.

3.6.4.1 Buzzer switch (See 4.8.12). The buzzer switch shall disconnect the buzzer for silent operation.

3.6.5 Signalling amplifier and oscillator (See 4.8.13). The signalling oscillator shall provide a frequency of 1600 HZ ± 2% over the specified operating temperature of the equipment. The signalling oscillator and amplifier shall provide a minimum of 9 volts RMS when terminated by a 430 ohm load.

3.6.6 Lead, Test CX-10354( )/PTM-7 (Orderwire Probe) (See 4.8.8). The CX-10354( )/PTM-7 shall provide for orderwire connection of Test Set, Telephone TS-1323( )/PTM-7 to Restorer, Pulse Form TD-206( )/G without interruption of PCM traffic.

3.6.7 Lead, Test CX-10355( )/PTM-7 (PCM Probe) (See 4.8.6). The CX-10355( )/PTM-7 shall be capable of detecting the PCM Signal when it exists at the Restorer, Pulse Form TD-206( )/G.

3.6.8 Talk-Listen (See 4.8.8). Order wire circuits shall provide clear, intelligible voice communication between Test Set, Telephone AN/PTM-7 and Multiplexer TD-204( )/U. For the test of 4.8.8, the 1 KHz component of the received orderwire signal shall be 50 mv P-P or greater and the 2.304 MHz component shall be 30 mv P-P or less. When signalling, a 1600 Hz ± 2% sine wave component shall be present on the output orderwire signal (Channel A). With orderwire switch OFF, no audio shall be present on the output orderwire signal. With the orderwire switch to "Talk", audio (voice) shall be present on the output orderwire signal. The 2.304 MHz orderwire component's P-P amplitude shall remain constant.

3.6.9 Switch position requirements.

3.6.9.1 OPERATION switch. The NORMAL position shall provide normal traffic flow in the cable system with the Test Set, Telephone TS-1323/PTM-7 connected at the TD-206( )/G restorer point. The LOOP-BACK position shall provide return of traffic back to the sending station.

3.6.9.2 DIRECTION switch. The DIRECTION switch shall select the transmission direction being monitored. The designations AB and BA shall refer to the A and B cable connectors.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.6.9.3 METER SELECT switch. The METER SELECT switch shall be capable of making operational connection with each of its eight positions.

3.6.9.3.1 Battery. This momentary contact position of the Meter Select Switch shall connect a load to the battery and cause the Test Meter to indicate in the green region for normal battery condition. For the test of 4.8.9.6, the Test Meter shall indicate 7½ or greater.

3.6.9.3.2 Probe. This position shall connect lead, test CX-10355( )/PTM-7 through the PCM detector (see 3.6.10) to the Test Meter. When tested in accordance with 4.8.6, a meter reading of 5 or greater shall be obtained. When tested in accordance with 4.8.13, a meter reading of 5 or greater shall indicate PCM is present and a reading of 2 or less shall indicate PCM is absent.

3.6.9.3.3 D.C. Volts. This position shall measure cable D.C. voltage at the TD-206( )/G input in selected direction (see 3.6.9.2). For the test of 4.8.9.1, the Test Meter shall indicate 5 + 1 minor division in AB direction and zero in BA direction with input voltage applied to connector 1. The Test Meter shall indicate 5 + 1 minor division in BA direction and zero in AB direction with input voltage applied to connector 8.

3.6.9.3.4 D.C. Amps. This position shall measure the cable current at the TD-206( )/G input in selected direction (see 3.6.9.2). A reading in the yellow region of the Test Meter shall indicate the normal condition of 38 milliamperes. For the test of 4.8.9.2, the Test Meter shall read between 6 and 9 in AB direction and zero in BA direction with S1 in position 1. The Test Meter shall read zero in AB direction and between 6 and 9 in BA direction with S1 in position 2.

3.6.9.3.5 Voltage Drop. This position shall measure D.C. voltage drop across the TD-206( )/G in selected direction (see 3.6.9.2). A reading in the lower portion of the green region shall indicate an operating TD-206( )/G while a reading in the upper portion of the green region (9 or greater) shall indicate a non-operating TD-206( )/G. For the test of 4.8.9.3, the Test Meter shall indicate as follows:

D.C. Voltage (+9.01v)	S1 Position	Direction	Test Meter Reading
9	1	AB	9 or greater
9	1	BA	zero
9	2	AB	zero
9	2	BA	9 or greater
7	1	AB	7 to 8
7	1	BA	zero
7	2	AB	zero
7	2	BA	7 to 8

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

3.6.9.3.6 PCM In. This position shall measure PCM signals at the TD-206( )/G input through the PCM detector (see 3.6.10) in selected direction (see 3.6.9.2). A reading in the green region of the Test Meter shall indicate the normal condition. For the test of 4.8.9.4, the Test Meter shall indicate 7 or greater.

3.6.9.3.7 PCM Out. This position shall measure PCM signals at the TD-206( )/G output through the PCM detector (see 3.6.10) in selected direction (see 3.8.9.2). A reading in the green region of the Test Meter shall indicate the normal condition. For the test of 4.8.9.5, the Test Meter shall indicate as follows:

Input Applied to Connector #	Direction	Operation	Test Meter Reading
7	AB	Normal	7 or greater
7	AB	Loopback	zero
2	BA	Normal	7 or greater
2	BA	Loopback	zero

3.6.9.3.8 Bridge (See 4.7.13 and 4.8.7). The Distance to Fault for either open or shorted (RES or CAP scale) conditions shall be indicated on the Distance to Fault indicator within ± 500 feet for a null reading of the Test Meter. The temperature correction chart shall be used with the RES scale to obtain the required accuracy. For the test of 4.8.7, a resistance of between 37 and 46 ohms shall cause a null with the Distance to Fault control set to 2 on the RES scale and a resistance of between 93 and 115 ohms shall cause a null with the Distance to Fault control set to 5 on the RES scale. A capacitance of between 0.046 and 0.056 uf shall cause a null with the Distance to Fault control set to 2 on the CAP scale and a capacitance of between 0.117 and 0.144 uf shall cause a null with the Distance to Fault control set to 5 on the CAP scale.

3.6.10 PCM Detector (See 4.8.14). A 25 millivolt peak-to-peak PCM input to the PCM Detector shall cause an indication in the yellow region of the Test Meter (6 to 8). A 2.2 volt peak-to-peak PCM Out signal shall cause the Test Meter to indicate 7 or greater.

3.6.11 Power (See 4.8.5). The equipment shall meet the requirements of 3.6 through 3.6.10 when powered from 8 dry cell batteries (BA-30). The BATTERY POWER switch shall apply power to the Test Set Telephone TS-1323/PTM-7 circuits. Closing the cover shall automatically return the BATTERY POWER switch to the OFF position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.6.12 Simplified instruction cards. One set of simplified instruction cards per SM-C-528098 shall be furnished with each equipment.

3.7 Sling, Carrying. One each sling, carrying per MIL-S-1698 shall be furnished with each equipment. (See 4.4).

3.8 Service conditions (See 4.7). The equipment shall meet the following service conditions.

3.8.1 Temperature.

3.8.1.1 Low. During and after testing as specified in 4.7.1.1, the equipment shall meet the requirements for the tests specified in 4.7.13.

3.8.1.2 High. During and after testing as specified in 4.7.1.2, the equipment shall meet the requirements for the tests specified in 4.7.13.

3.8.2 Humidity. The equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements for the test of 4.7.13 during the operating times specified in 4.7.2.

3.8.3 Low Pressure (Altitude). During and after testing as specified in 4.7.3, the equipment shall meet the requirements for the tests specified in 4.7.13.

3.8.4 Vibration, Bounce and Shock.

3.8.4.1 Vibration, internal. During and after testing as specified in 4.7.4, the equipment shall meet the requirements for the tests specified in 4.7.13.

3.8.4.2 Shock (bounce). The equipment shall meet the requirements for the tests specified in 4.7.13 after subjection to the test specified in 4.7.5, LOAD: 10G.

3.8.4.3 Shock (bench handling). The equipment shall meet the requirements for the tests specified in 4.7.13 after subjection to the test specified in 4.7.6.

3.8.5 Rain. The equipment shall meet the requirements for the test specified in 4.7.13 after subjection to the test specified in 4.7.9. There shall be no evidence of water leakage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.8.6 Shock (drop). The equipment shall meet the requirements for the tests specified in 4.7.13 after subjection to the test specified in 4.7.7. Exterior paint damage shall not be cause for rejection.

3.8.7 Leakage (Immersion). The equipment shall show no evidence of water leakage after subjection to the test specified in 4.7.8.

3.8.8 Salt fog. After the salt fog test of 4.7.10, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation (see 6.3) of metal. Electrical operation is not required.

3.8.9 Dust (fine sand). After testing as specified in 4.7.11, the equipment shall meet the requirements for the tests specified in 4.7.13. Except for exterior paint damage on the case or front panel, the equipment shall exhibit no electrical or mechanical damage.

3.8.10 Fungus. After the fungus test of 4.7.12, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3) as evidenced by growth colonization (see 6.3) (which includes branching (see 6.3) and sporulation (see 6.3)) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3) shall not be included in this evaluation.

3.9 Preconditioning.

3.9.1 Electrical Preconditioning (See 4.6.2). Prior to submission to the specified Quality Conformance Inspection, each assembled Test Set, Telephone AN/PTM-7 shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process.

3.9.2 Bounce Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1. Each unit shall be subjected to this process.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.10 Interchangeability (See 4.10). Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.11 Electromagnetic Compatibility (See 4.9). The AN/PTM-7 shall meet the Electormagnetic Interference requirements specified in MIL-STD-461A, Notice 4, Table A-1 for Class CE equipment except that only tests RE02, RE02.1, and RS03.2 are applicable. Tests and test methods shall be as specified in MIL-STD-462, Notice 3.

3.12 Printed wiring assemblies (See 4.4). Printed wiring assemblies shall be in accordance with the applicable paragraph in MIL-P-11268 and shall be conformally coated in accordance with MIL-STD-275 with a coating material that conforms to MIL-I-46058, type ER (Epoxy).

3.13 Operational inspection. Each fabricated and tested Test Set, Telephone AN/PTM-7( ) shall be subjected to a final inspection in accordance with 4.8.5 after all other inspection and tests are completed to determine that the equipment is operable prior to packaging.

3.14 Leads, test. Test leads CX-10354( )/PTM-7 and CX-10355( )/PTM-7 shall be tested and meet the requirements of MIL-C-3885 (See 4.4).

3.15 Panel assemblies (Electrical Tests)(See 4.14). The AN/PTM-7( ) and panel assemblies listed below shall meet the requirements of their respective test requirement drawings as specified in 4.14.

Test Set, Telephone AN/PTM-7( ) (TRS Test and Alignment Requirements)	SM-B-528048
Panel Assembly 8A1	SM-B-528165
Panel Assembly 8A2	SM-B-528166
Panel Assembly 8A3	SM-B-528167

3.16 Safety Factors (See 4.11).

3.16.1 Radioactive material shall not be employed in the AN/PTM-7( ).

3.16.2 Finished items shall not contain any raw, sharp, or rough edges on any parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.17 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-Tapping screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspections shall be made available by the contractor.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery (see 4.3)).

b. Inspection covered by subsidiary documents (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish five First Article units of the complete Test Set, Telephone AN/PTM-7( ).

4.3.2 First Article Inspection. The First Article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First Article Data. The First Article Test Plan and Test Report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary document as part of the inspection of equipment before parparation for delivery:

ITEM	WHERE REQUIRED
Finish	3.4
Marking	3.5
Parts, Materials and Processes	3.3
Headset-Microphone	3.2.1
Sling, Carrying	3.7
Printed Wiring Assemblies	3.12
Leads, Test	3.14

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

TABLE I. ORDER OF FIRST ARTICLE INSPECTION

Inspections	Reqt Para	Test Para	Number of Samples - Five				
			Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
1. Inspections covered by subsidiary documents		4.4	X	X	X	X	X
2. Bounce preconditioning	3.9.2	4.6.1	X	X	X	X	X
3. Electrical preconditioning	3.9.1	4.6.2	X	X	X	X	X
4. Group A Inspection	SEE TABLE II		X	X	X	X	X
5. Group B Inspection	SEE TABLE III		X	X	X	X	X
6. Group C Inspection							
Low temperature	3.8.1.1	4.7.1.1		1			
High temperature	3.8.1.2	4.7.1.2		2			
Humidity	3.8.2	4.7.2	3				
Low pressure (Altitude)	3.8.3	4.7.3	2				
Vibration	3.8.4.1	4.7.4			1		
Shock (Bounce)	3.8.4.2	4.7.5			2		
Shock (Bench Handling)	3.8.4.3	4.7.6			3		
Fungus (see Note C)	3.8.10	4.7.12				1	
Rain	3.8.5	4.7.9	1				
Salt Fog (see Note C)	3.8.8	4.7.10		4			
Dust (Fine Sand)	3.8.9	4.7.11		3			
EMI	3.11	4.9					1
Shock (Drop)	3.8.6	4.7.7					2
Leakage (Immersion)	3.8.7	4.7.8			4		

NOTES:

NOTE A: The Inspections 1 through 5 shall be performed in the order shown on all First Article units before subjecting these units to any other inspection requirements specified in the Table.

NOTE B: The numbers in the unit columns in the Table covered by Inspection 6 specify the order of inspections for the indicated units.

NOTE C: The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent tests.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.1 Group A Inspection. Each unit (and each subassembly identified in 3.15) on contract or purchase order shall be inspected for conformance to the inspections specified in Table II (Manufacturing Inspection). Units shall be comprised of subassemblies which have been subjected to and passed the Manufacturing Inspection and Quality Assurance Verification cited herein. Discrete lots for Quality Assurance Verification shall be formed from units and subassemblies that pass the Manufacturing Inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification).

TABLE II - GROUP A INSPECTION

Inspection	Req Para	Ref Fig	Insp Para	AQL	
				*Major	Minor
Visual-Mechanical	3.17		4.11	1%	4%
Simplified Instruction Cards	3.6.12,3.17		4.11	1%	4%
Sling, Carrying	3.7, 3.17		4.11	1%	4%
TS-1323( )/PTM-7	3.2, 3.17		4.11	1%	4%
Panel Assembly 8A1	3.2, 3.17		4.11	1%	4%
Panel Assembly 8A2	3.2, 3.17		4.11	1%	4%
Panel Assembly 8A3	3.2, 3.17		4.11	1%	4%
AN/PTM-7( ) Case and Cover	3.17		4.11	1%	4%
Electrical Performance (Including Spare Panel Assy's)					
Panel Assembly 8A1	3.15		4.14	1%	N/A
Panel Assembly 8A2	3.15		4.14	1%	N/A
Panel Assembly 8A3	3.15		4.14	1%	N/A
Test Set, Telephone AN/PTM-7( )	3.15		4.14	1%	N/A
Electrical Tests					
Lead, Test CX-10354( )/PTM-7	3.6.6,3.6.8	4	4.8.8	1%	N/A
Lead, Test CX-10355( )/PTM-7	3.6.7, 3.6.9.3.2	6	4.8.6	1%	
Bridge Test	3.6.9.3.8	5	4.8.7	1%	N/A
Function Test	3.6.9.1, 3.6.9.2 , 3.6.9.3.1 thru 3.6.9.3.7	7	4.8.9 thru 4.8.9.6	1%	N/A
Operational Test	3.13	3	4.8.5	1%	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and the special inspection levels of Table I of MIL-STD-105.

TABLE III - GROUP B INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Interchangeability	3.10		4.10
Receiver Amplifier	3.6.2	8	4.8.10
Transmit Amplifier	3.6.3	9	4.8.11
Ringing Detector	3.6.4	10	4.8.12
Signal Amp & Oscillator	3.6.5	11,13	4.8.13
PCM Detector	3.6.10	12	4.8.14

4.5.2.1 Group B sampling plan. The Group B AQL shall be 6.5 percent defective and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table IV, and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

TABLE IV
GROUP C INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Sub-Group 1			
Temperature	3.8.1	1	
Low	3.8.1.1		4.7.1.1
High	3.8.1.2		4.7.1.2
Moisture Resistance (Humidity)	3.8.2		4.7.2 (notes 1 & 2)
Leakage (Immersion)	3.8.7	1	4.7.8
Sub-Group 2			
Vibration (internal)	3.8.4.1	1	4.7.4
Shock (bounce)	3.8.4.2	1	4.7.5 (note 1)
Shock (bench handling)	3.8.4.3	1	4.7.6
Rain	3.8.5	1	4.7.9
Shock (drop)	3.8.6	1	4.7.7 (note 1)
Sub-Group 3			
Salt Fog	3.8.8		4.7.10 (See notes 1 & 2)
Fungus	3.8.10		4.7.12 (See notes 1 & 2)
Dust (fine sand)	3.8.9		4.7.11
Low Pressure (altitude)	3.8.3	1	4.7.3
EMI	3.11	15	4.9

NOTE 1: Unless otherwise specified in the contract or order a different unit shall be used for each of these tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished if necessary, before proceeding with subsequent tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every month. Both units shall be subjected to temperature and humidity testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every three months. The damage inducing tests shall not be performed on more than one unit.

4.5.3.1.3 Subgroup 3. For this subgroup, three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every sixth month. The damage inducing tests shall not be performed on more than one unit.

4.5.5 Re-inspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible and known damage.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government except that immersion and rain shall be done after the drop test, where available.

4.5.4 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.6 Preconditioning (see 3.9).

4.6.1 Bounce preconditioning (3.9). Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the Package Tester Type 1000-SC as made by the L.A.B. Corporation, Skaneateles, New York or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute prior to selection of units for inspection. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. The requirements of 3.9 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.2 Electrical preconditioning (see 3.9). During electrical burn-in, the equipment shall be connected per Figure 14. There shall be a 25MV peak-to-peak 2.3OMHZ signal applied to the PCM Probe Input. Test level B of MIL-STD-781 shall apply for Temperature and Vibration Requirements. Upon completion of this preconditioning, the unit shall be tested in accordance with 4.8.5 (Operational test). If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) shall be replaced and the preconditioning cycle re-instituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.9 shall be met.

4.6.2.1 Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning, may be repaired with an electrically preconditioned part and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.6.2.2 All electrically preconditioned panel assemblies and chassis shall be physically identified by a preconditioned stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group 3 requirements of MIL-M-13231.

4.7 Service condition test. The following service condition tests shall be performed. Since more than one unit is available, some tests may be run concurrently with Government approval by the contracting officer or his designated representative. After completion of the Fungus and Salt Fog tests, the equipment must be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.7.1 Temperature.

4.7.1.1 Low temperature (see 3.8.1.1). The equipment shall be tested in accordance with Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be minus 65 degrees F. Storage time shall be 72 hours. Operation temperature shall be minus 25 + 0 -5 degrees F. At operation temperature, the test shall start within 5 minutes after power application to the unit. Prior to testing, the equipment shall meet the requirements for the test specified in 4.7.13. Performance checks shall be made during tests as specified in 4.7.13 and Test Method 502.1. Two power supplies, each current limited to one amp, shall be used to supply +4.50V and -5.25V in lieu of batteries.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.7.1.2 High temperature (see 3.8.1.2). The equipment shall be tested in accordance with Test Method 501.1, Procedure II of MIL-STD-810. Operating temperature shall be plus 125 degrees F. Prior to testing, the equipment shall meet the requirements for the test specified in 4.7.13. Performance check shall be made during tests as specified in 4.7.13 and Test Method 501.1.

4.7.2 Moisture resistance test (humidity) (see 3.8.2). The equipment shall be subjected to the humidity test specified in Test Method 507.1, Procedure II of MIL-STD-810 with test measurements during Step 4, and during the 24 to 28 hour period and last 5 hour period of each cycle in Step 6. Inability of the equipment to meet the requirements of 3.8.2 shall constitute failure of this test.

4.7.3 Low pressure (Altitude) (see 3.8.3). The equipment shall be subjected to the following test:

a. The equipment shall be placed in the test chamber. Prior to testing, the equipment shall meet the requirements for the tests specified in 4.7.13 except that auxiliary test equipment need not be inclosed in the chamber. A temperature of $73^{\circ} \pm 18^{\circ}$F shall be maintained during the entire test.

b. Reduce the chamber internal pressure to 87.5 mm of HG (3.44 in of Hg or 50,000 feet above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

c. Increase the chamber pressure to 523 mm of Hg (20.6 in of Hg or 10,000 feet above sea level) and then perform the test of 4.7.13 (do not signal from the AN/PTM-7) and record results.

d. With the equipment not operating, return the chamber to standard ambient conditions and perform the test of 4.7.13 and visually inspect the equipment.

e. During and after the test, the equipment shall meet the requirements for the test of 4.7.13 and shall not exhibit any electrical or mechanical degradation which could in any manner prevent the equipment from meeting any requirements of this specification.

4.7.4 Vibration, Internal (see 3.8.4.1). The equipment shall be subjected to the following test:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

a. The equipment shall be secured to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that the vibration within the test item can be observed and measured.

b. The equipment shall be vibrated successively in three mutually perpendicular directions over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall not be less than 0.030 inch. In each of the three directions, the frequency shall be changing in steps of 1 Hz and maintained for at least 10 seconds. Vibration amplitudes shall be measured by optical or other means, provided the vibration of the test item is not affected by the measurement.

c. The equipment shall have no resonance in the frequency range of 10 to 55 Hz that exceeds twice the amplitude of the applied vibration.

4.7.5 Shock test (bounce) (see 3.8.4.2). The equipment shall be tested in accordance with Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Prior to and after testing, the equipment shall meet the requirements for the test specified in 4.7.13 except that the equipment need not be inclosed in a chamber.

4.7.6 Shock test (bench handling) (see 3.8.4.3). The equipment shall be tested in accordance with Test Method 516.2, Procedure V of MIL-STD-810. Prior to and after testing, the equipment shall meet the requirements for the test specified in 4.7.13 except that the equipment need not be inclosed in a chamber.

4.7.7 Shock test (drop) (see 3.8.6). The equipment shall be tested in accordance with Test Method 516.2, Procedure II of MIL-STD-810. Height of drop shall be 2 feet. Number of drops shall be one on each corner (total of eight). Prior to and after testing, the equipment shall meet the requirements for the test specified in 4.7.13 except that the equipment need not be inclosed in a chamber.

4.7.8 Immersion test (see 3.8.7). The equipment as prepared for field transporation shall be tested in accordance with Test Method 512.1, Procedure I of MIL-STD-810. Equipment depth shall be changed to 12 + 0.5 -0 inches of water.

4.7.9 Rain Test (see 3.8.5). The equipment shall be subjected to the following test.

a. Condition for 4 hours at 77° ± 5°F and 40 to 60 percent relative humidity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. Test in accordance with 4.7.13.

c. Place the equipment, non-operating, but in its normal operating position with top cover off in a test chamber maintained at 77°F ± 5°F as shown in Figure 1. Allow simulated rainfall to fall on exposed surfaces of the component for 1 hour. The water shall be at a temperature of 65°F ± 5°F and shall fall freely at a rate of 2 inches per hour.

d. With the rainfall continuing at the 2 inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. Maintain this condition for one hour.

e. Rotate the equipment 90 degrees horizontally to expose a second side of the equipment to the wind and rain. Maintain this condition for one hour.

f. Continue as in e. above until all four sides of the equipment have been exposed to the wind and rain, each side for one hour.

g. Immediately after completion of the foregoing, repeat the measurement specified in b. above without removal from the chamber but with the rain and wind discontinued. Complete the measurements within one hour. The measurements shall meet specified performance.

4.7.10 Salt fog (see 3.8.8). The equipment (*including accessories*), in its operating configuration, with the top cover off shall be tested in accordance with Test Method 509.1 of MIL-STD-810. At the conclusion of the test, the equipment shall be examined for evidence of visible corrosion. No rust shall be imposed on the equipment by the test facility. The equipment will not be operated during this test but shall pass all specification requirements when tested in accordance with 4.8.5 before and after the salt fog test.

4.7.11 Dust (Fine sand) (See 3.8.9). The equipment, in its operating configuration, shall be subjected to Test Method 510.1, Procedure I of MIL-STD-810. At the conclusion of this test, the equipment shall be operated and shall meet full specified performance when tested in accordance with 4.8.5. The equipment shall not be operated during the test. Prior to testing, the equipment shall meet the requirements for the test specified in 4.8.5.

4.7.12 Fungus (see 3.8.10). The equipment shall be tested in accordance with Test Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (See 6.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment, together with cases and accessories, is permitted 72 hours prior to the fungus test. The equipment shall be exposed with the chassis removed from the case. Inability of the equipment to meet full specified performance when tested in accordance with 4.8.5 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the currentversion before use.

4.7.13 Chamber tests (service condition/operating and non-operating).

a. Connect equipment per Figure 1.

b. Align equipment for normal operation.

c. Using Headset-Microphone H-91( )/U, perform talk and listen tests over orderwire circuits between terminals and between AN/PTM-7( ) and terminals.

d. Signal over orderwire circuits from each terminal and from AN/PTM-7.

e. Note meter indication in each of the following METER SELECT switch position: BATTERY, DC VOLTS, DC AMPS, VOLT DROP, PCM IN and PCM OUT for both AB and BA direction.

f. Interrupt traffic in AB direction (remove TIM IN cable from Multiplexer TD-204( )/U "A") and repeat e. above.

g. Operate OPERATION switch to LOOP-BACK position and note proper operation of Multiplexer TD-204( )/U and TD-660( )/U.

h. Connect equipment per Figure 2.

i. With switch SI in position I determine distance to cable open circuit using BRIDGE circuit.

j. With switch SI in position 2 determine distance to cable short circuit using BRIDGE circuit.

k. The requirements of 3.6.1, 3.6.3, 3.6.6, 3.6.7, 3.6.8, 3.6.9.1, 3.6.9.2, 3.6.9.3, 3.6.9.3.1, 3.6.9.3.3, 3.6.9.3.4, 3.6.9.3.5, 3.6.9.3.6, 3.6.9.3.7, and 3.6.9.3.8 shall be met.

4.8 Electrical tests.

4.8.1 Standard conditions. Unless otherwise specified, all tests shall be performed at 12 volts DC nominal supplied by eight dry cell batteries BA30 installed in the unit under test. Tests shall be conducted at room ambient temperature, pressure and relative humidity per MIL-STD-810 unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

4.8.2 Alignment for normal operation. No alignment is required for Test Set, Telephone AN/PTM-7 when tested in accordance with this specification.

4.8.3 Standard test equipment. Standard test equipment is indicated in test figures of this specification and the Test Requirement Specification (TRS) which is part of the ECOM Drawing SC-DL-528168 by manufacturer and model number. In all cases equivalent substitutes may be used with prior approval of the contracting officer's technical representative (COTR). Unless otherwise specified, standard test equipment shall be provided by the contractor subject to the approval of the COTR.

4.8.4 Special test equipment. Special test equipment is indicated by name and number in test figures of this specification and the TRS which are part of the ECOM Drawings SC-DL-528168. Unless otherwise specified special test equipment shall be provided by the contractor and equivalent substitutes may be used subject to the approval of the Government. Cable assemblies shall be provided by the contractor in the quantities necessary and with cable lengths as required to accomplish the testing herein. The contractor need not supply the CX-11230( )/G cable. The cable assemblies are as follows:

a. Cable Assembly, Power Electrical, CX-11727/G - The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-525135.

b. Cable Assembly, Radio Frequency, CG-1040( )/U - The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531012.

c. Cable Assembly, Radio Frequency, CG-2437/TCC - The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531016.

d. Cable Assembly, Radio Frequency, CG-2438/TCC - The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531017.

e. Cable Assembly, Adaptor CX-10734/G - The cable assembly shall be constructed in accordance with and meet the requirements of DL-SC-A-339551 and Military Specification MIL-C-55583(EL).

4.8.5 Operational Test (see 3.14).

a. Connect equipment per Figure 3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the currentversion before use.

b. Signal, talk and listen over orderwire from Test Set to both Multiplexers TD-204( )/U.

c. Check for proper indication on TEST METER with the METER SELECT switch in the BATTERY, DC VOLTS, DC AMPS, VOLT DROP, PCM IN, and PCM OUT positions.

d. Pull battery on-off switch to on position. Close top cover.

e. The requirements of 3.6.1, 3.6.4, 3.6.8, 3.6.9.3, 3.6.9.3.1, and 3.6.9.3.3 through 3.6.9.3.7 and 3.6.11 shall be met.

4.8.6 Probe Test - PCM.

a. Connect equipment per Figure 4.

b. Perform battery check.

c. With HFO set for 2.304 MHz, 300 mv P-P, insert CX-10355 probe tip into probe well and set meter select switch (S3) to "Probe."

d. Record meter reading.

e. The requirements of 3.6.9.3.2 and 3.6.7 shall be met.

4.8.7 Bridge Test.

a. Connect equipment per Figure 5.

b. Align equipment for resistance reading.

c. With Switch S1 in position 1 and "Distance to Fault" control (R68) set to 2 on resistor scale, adjust decade resistor for null on test meter (M1) and record resistance.

d. Repeat c. for R68 set to 5 on resistor scale.

e. Align equipment for capacitance reading.

f. With Switch S1 in position 2 and "Distance to Fault" control (R68) set to 2 on cap scale, adjust decade capacitor for null on test meter (M1) and record capacitance.

g. Repeat f. for R68 set to 5 on cap scale.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

h. The requirements of 3.6.9.3.8 shall be met.

4.8.8 Probe Test (Orderwire).

a. Connect equipment per Figure 6.

b. Perform battery check.

c. With S2 open, adjust HFO for a 200 mv P-P, 2.304 MHz square wave on CH.A.

d. Close S2 and adjust LFO for a 300 mv P-P, 1.0 KHz sine wave on CH.A.

e. With Sw 1 in position 1, measure P-P amplitude of CH.B. with time base of 200 us (1 KHz component).

f. Repeat e. with time base of 0.1 us (2.304 MHz component).

g. Repeat e. and f. for Sw 1 in position 2.

h. Repeat e. and f. for Sw 1 in position 3.

i. Repeat e. and f. for Sw 1 in position 4.

j. With Sw 1 open, operate orderwire switch momentarily to "SIG" position and measure amplitude of 1600 Hz component on CH.A.

k. With S2 open, operate orderwire switch to "OFF" position and talk into Headset H1. Note absence of low frequency component on CH.A.

l. With S2 open, operate orderwire switch to "Talk" position and talk into headset. Note presence of low frequency component on CH.A.

m. The requirements of 3.6.8 shall be met.

4.8.9 Function Test.

4.8.9.1 D.C. Volts.

a. Connect equipment per Figure 7A.

b. Perform Battery check.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

c. With meter select switch (S3) in "D.C. Volts" position and "Direction" switch in AB position, apply 500 ± 5 V DC to connector 1.

d. Repeat c. with "Direction" switch in BA position.

e. Repeat c. and d. with voltage applied to connector 8.

f. The requirements of 3.6.9.3.3 shall be met.

4.8.9.2 D.C. Amps.

a. Connect equipment per Figure 7B.

b. Perform battery check.

c. With S1 in position 1 and direction switch to AB, apply 3.5 ± 0.01 V DC as shown and record test meter indication.

d. Repeat c. with direction switch in BA position.

e. Repeat c. and d. with S1 in position 2.

f. The requirements of 3.6.9.3.4 shall be met.

4.8.9.3 Voltage Drop.

a. Connect equipment per Figure 7C.

b. Perform Battery Check.

c. With S1 in position 1 and direction switch to AB apply 7 ± 0.01 V DC as shown.

d. Repeat c. with direction switch to BA.

e. Repeat c. and d. with S1 in position 2.

f. Repeat c., d., and e., for an input voltage of 9 ± 0.01 V DC.

g. The requirements of 3.6.9.3.5 and 3.6.9.2 shall be met.

4.8.9.4 PCM IN.

a. Connect equipment per Figure 7D.

b. Perform battery check.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

c. With direction switch to AB, apply a 2.304 MHz, 25 mv P-P square wave to connector 3 via cable CX-10354 and record test meter indication.

d. Repeat c. with direction switch to BA and signal applied to connector 6.

e. The requirements of 3.6.9.3.6 shall be met.

4.8.9.5 PCM OUT.

a. Connect equipment per Figure 7D.

b. Perform battery check.

c. With direction switch to AB and operation switch to normal, apply a 1.9v P-P, 2.304 MHz square wave to connector 7 via CX-10354 lead.

d. Repeat c. for operation switch to loopback.

e. Repeat c. and d. for direction switch to BA and signal applied to connector 2.

f. The requirements of 3.6.9.3.7 and 3.6.9.2 shall be met.

4.8.9.6 Battery Test.

a. Connect equipment per Figure 7E.

b. With unit turned on, momentarily place meter select switch to "Battery" position and record reading on test meter.

c. The requirements of 3.6.9.3.1 shall be met.

4.8.10 Receive Amplifier Test.

a. Connect equipment per Figure 8.

b. Align equipment for normal operation.

c. Apply 300 millivolt RMS 1 KHZ signal between E1 and E2.

d. Measure output at receive amplifier output.

e. Repeat c. and d. above for 1.0v, 2v, 4v, 8v, and 16v input.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

f. The requirements of 3.6.2 shall be met.

4.8.11 Transmit Amplifier Test.

a. Connect the equipment per Figure 9.

b. Align equipment for normal operation.

c. Place TALK-OFF-SIGNAL switch in TALK position.

d. Apply 100 millivolt RMS 1 KHZ signal to input of HEADSET connector.

e. Measure output at J23.

f. The requirements of 3.6.3 shall be met.

4.8.12 Ringing Detector Test.

a. Connect equipment per Figure 10.

b. Align equipment for normal operation.

c. Apply 300 millivolt RMS 1600 HZ signal to E1.

d. Determine proper operation of signal lamp and buzzer. Check for proper operation of buzzer ON-OFF switch.

e. Repeat c. and d. for 10v and 16v signals.

f. Repeat c., d., and e. for signals of 1568 and 1632 HZ.

g. The requirements of 3.6.4 and 3.6.4.1 shall be met.

4.8.13 Signalling Amplifier and Oscillator Test.

a. Connect equipment per Figure 11.

b. Align equipment for normal operation.

c. Operate TALK-OFF-SIGNAL switch to the SIGNAL position and measure level and frequency of output at E1.

d. Connect equipment per Figure 13.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

e. Repeat c. above.

f. The requirements of 3.6.5 shall be met.

4.8.14 PCM Detector Test.

a. Connect equipment per Figure 12.

b. Align equipment for normal operation.

c. Apply a 25 mv peak-to-peak 2.304 MHZ signal to the PCM PROBE INPUT.

d. Set METER SELECT switch to PROBE and note indication of TEST METER.

e. Apply 2.2 volts peak-to-peak 2.304 MHZ signal to J5.

f. Set METER SELECT switch to PCM OUT position and note indication on TEST METER.

g. The requirement of 3.6.10 shall be met.

4.9 Electromagnetic compatibility. The AN/PTM-7 shall be tested for compliance to the requirements of MIL-STD-461A, Notice 4, Table A-I for CE equipment. Only tests REO2, REO2.1, and RSO3.2 shall be performed. Tests and test methods shall be per MIL-STD-462, Notice 3. All electromagnetic compatibility tests shall be performed with the AN/PTM-7 simulating normal operation. The frequency range for all tests shall be 14 KHz to 1000 MHz. The test procedure shall be as follows:

a. Connect equipment per Figure 15.

b. Perform radiated emission tests on the equipment in accordance with MIL-STD-462 and the approved EMC/EMI test plan.

c. Perform radiated susceptibility tests on the equipment in accordance with MIL-STD-462 and the approved EMC/EMI test plan while performing the following test: Operational Test per 4.8.5.

d. The requirements of 3.11 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.10 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the mechanical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts of assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors. Locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252 for conformance with MIL-STD-454 and the requirements of 3.3 and 3.17.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.13 Rough handling test (preparation for delivery). When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2.e), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling: Operational Test of 4.8.5.

4.14 Electrical Subassembly Test (see 3.15). Telephone Test Set AN/PTM-7( ) and panel assemblies, including running spare panel assemblies shall be tested in accordance with their respective test requirement specification drawings listed in 3.15.

5. Preparation For Delivery.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2c).

5.1.1 Level A.

5.1.1.1 Cleaning. Test Set, Telephone AN/PTM-7 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Test Set, Telephone AN/PTM-7 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Stowage. Stow the components of the Test Set, Telephone AN/PTM-7, with the spare lamp, within the spaces provided in the cover of the carrying case and fill all voids with paperboard conforming to PPP-P-291, Type I, to prevent movement. Secure cover with fasteners provided. Close cover of carrying case and secure with fasteners provided.

5.1.1.4.3 Test Set, Telephone AN/PTM-7 shall be individually packaged method III as follows: Cushion each test set on all surfaces with pads or cells or both fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather resistant, variety SW, Grade W6c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W6c. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or Polyurethane foam conforming to MIL-P-26514. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with appendix of the box specification.

5.1.2 Level C. Test Set, Telephone AN/PTM-7, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in-transit storage.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the currentversion before use.

5.2 Packing. Packing shall be Level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads, during shipments and storate without damaging the containers or its contents (see 6.1).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Test Set, Telephone AN/PTM-7, packaged as specified in 5.1 shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodated marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type 1, Finish A or shrink film.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2, or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, Finish A.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Test Set, Telephone AN/PTM-7, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Test Set, Telephone AN/PTM-7, packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to MIL-P-15011 or MIL-P-3938. Closure shall be in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape bonding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Test Set, Telephone AN/PTM-7 packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Test Set, Telephone AN/PTM-7 packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1, and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Test Set, Telephone AN/PTM-7( ) is a portable lineman's test set intended primarily to check the performance of unattended binary cable repeaters, to locate cable faults in the mile of unrepeatered cable span of cable assemblies, special purpose, Electrical CX-11230( )/G and to provide communications with attended terminal pcints located up to a maximum distance of 40 miles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required.

c. Levels of preservation, packaging, and packing (see Section 5).

d. Marking and shipping of samples.

e. When rough handling tests is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

f. Place of final inspection.

g. First Article inspection.

h. Technical literature required.

i. Quantity of tools and running spare parts required.

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.3.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipment. Initial shipment of end items will be contingent only upon successful completion of all Group C tests (see 4.5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

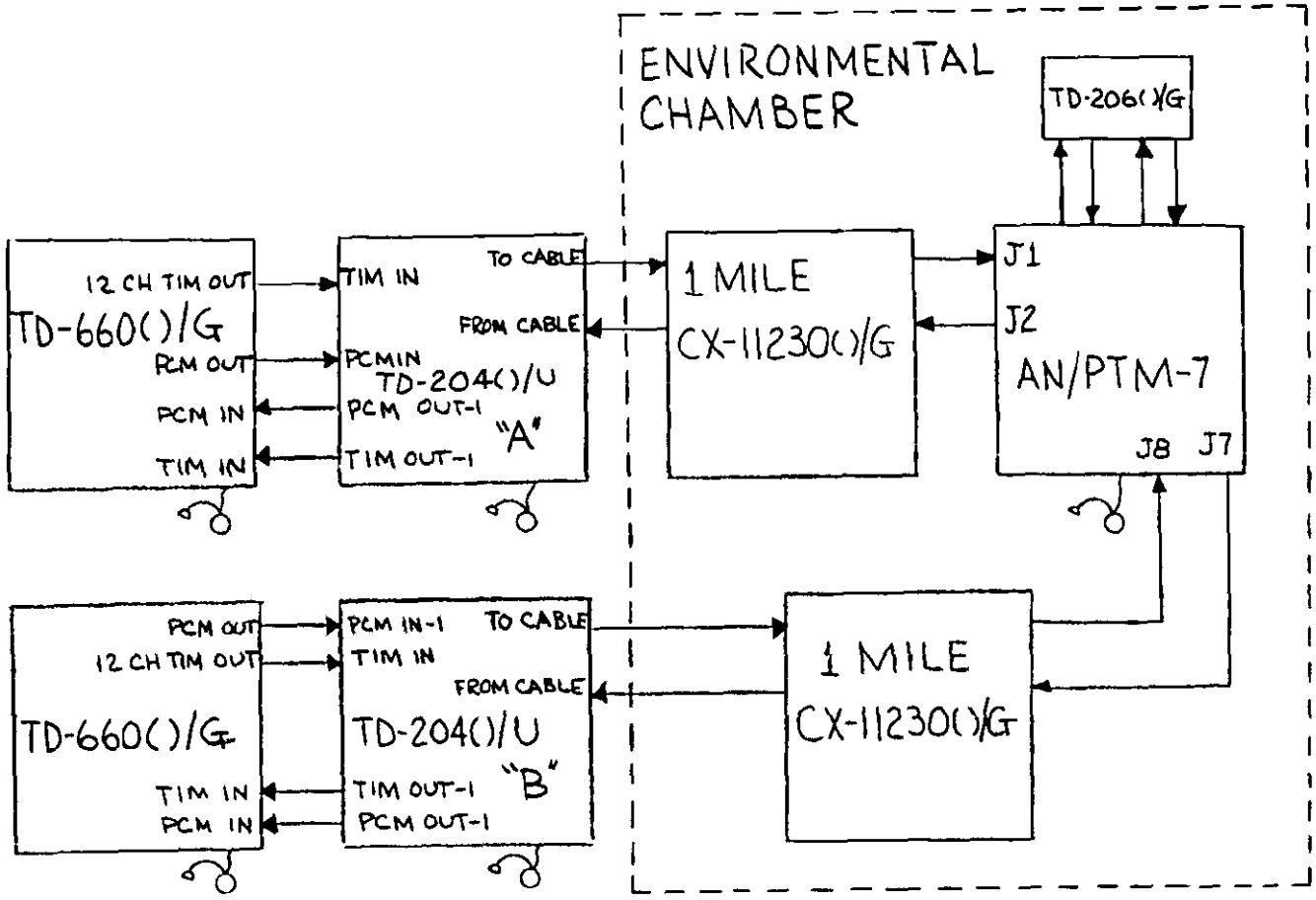
6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example: AN/PTM-7W. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
Army-EL

Preparing Activity
Army-EL
Project Number 6625-A640

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.



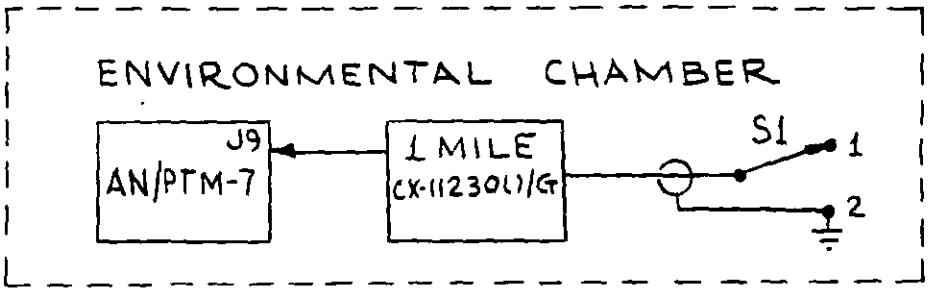
NOTES:

1. 115 VAC 50-60 HZ POWER TO TD-660 AND TD-204 UNITS VIA CX-11727/G.
2. CX-10734( )/G ADAPTOR BETWEEN TD-204 AND CX-11230 AND AN/PTM-7 AND CX-11230
3. ∿ DENOTES HEADSET H-91( )/U

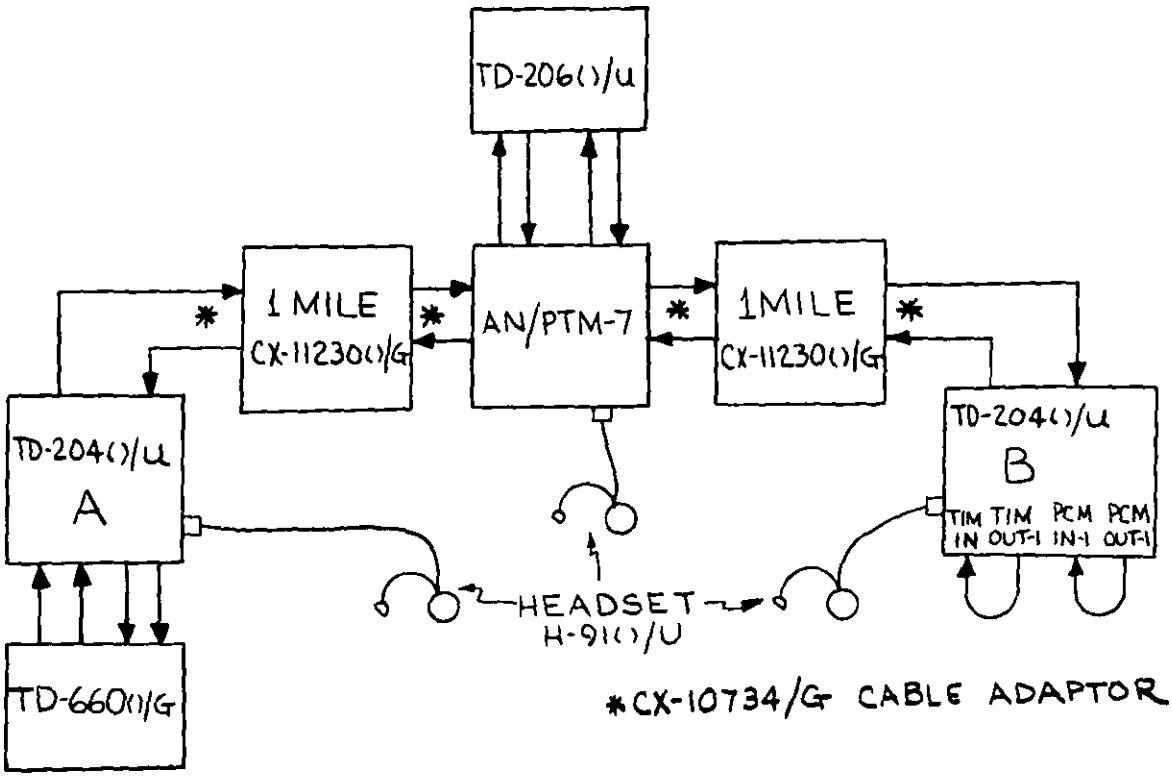
CHAMBER TESTS

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

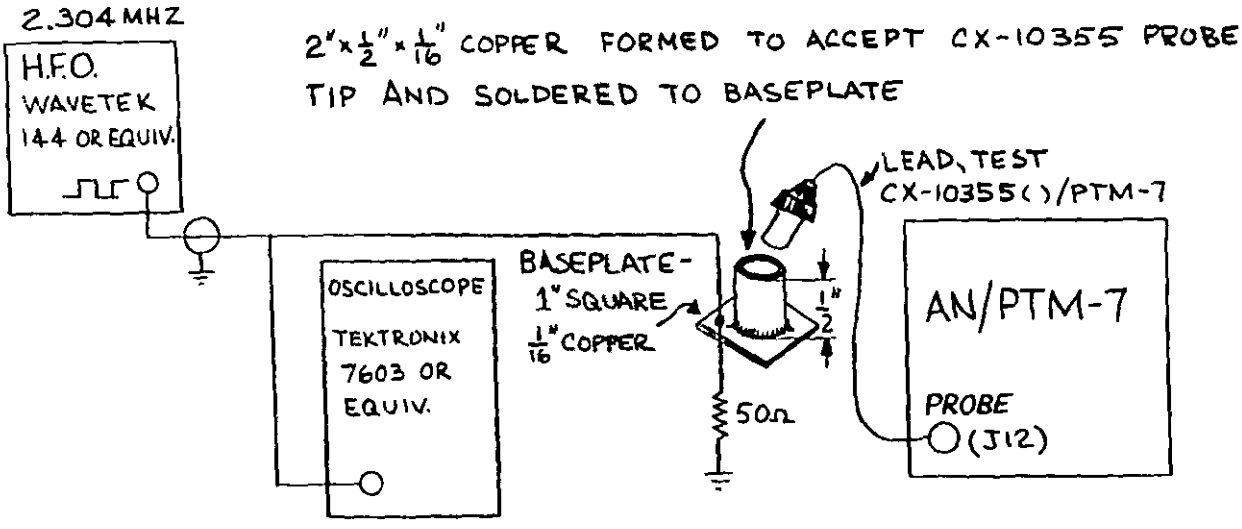


CHAMBER TESTS
FIGURE 2



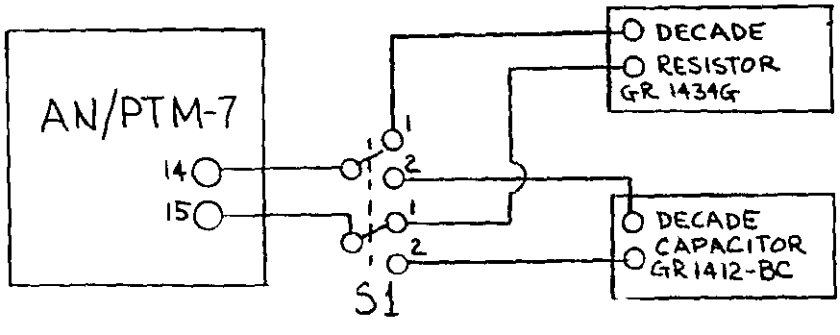
OPERATIONAL TEST
FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.



PROBE TEST - PCM

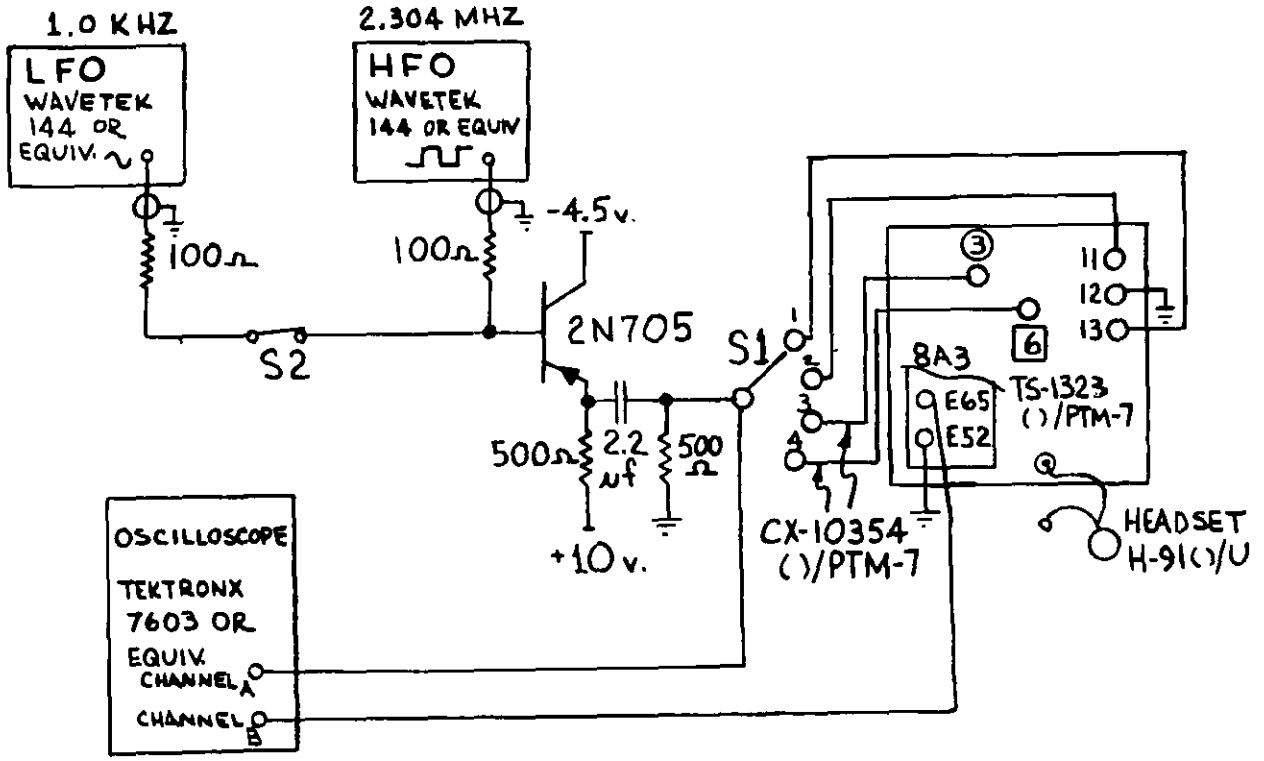
FIGURE 4



BRIDGE TEST

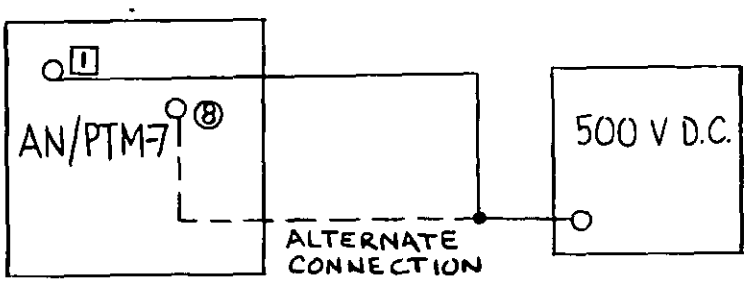
FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.



PROBE TEST - ORDERWIRE

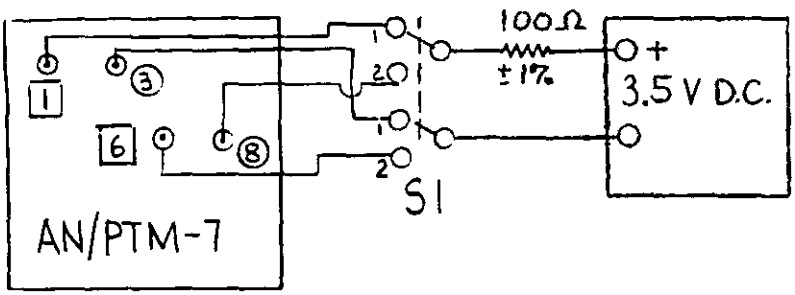
FIGURE 6



FUNCTION TEST - D.C. VOLTS

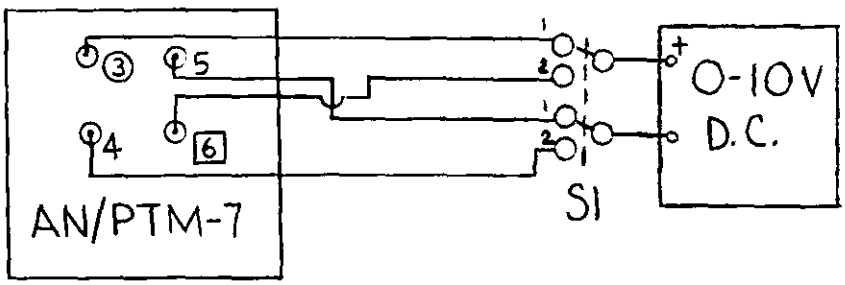
FIGURE 7A

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.



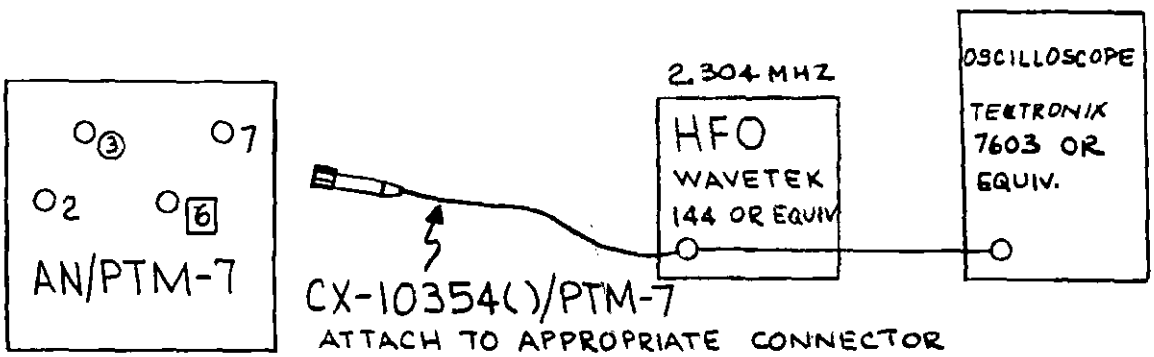
FUNCTION TEST- D.C. AMPS

FIGURE 7B



FUNCTION TEST- VOLTAGE DROP

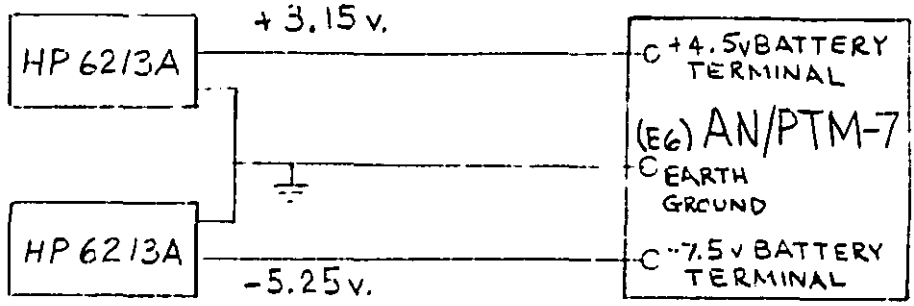
FIGURE 7C.



FUNCTION TEST-PCM IN & PCM OUT

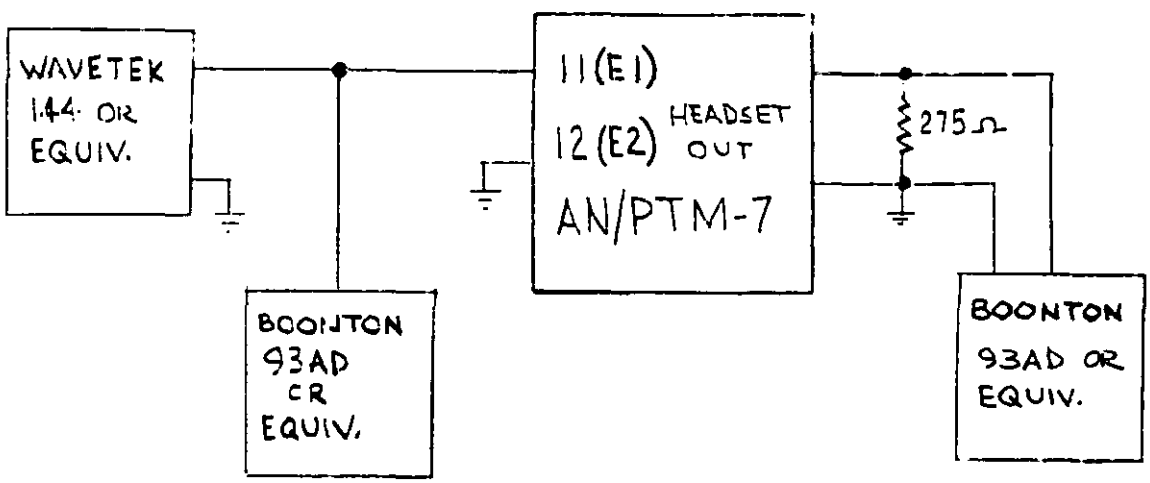
FIGURE 7D

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.



BATTERY TEST

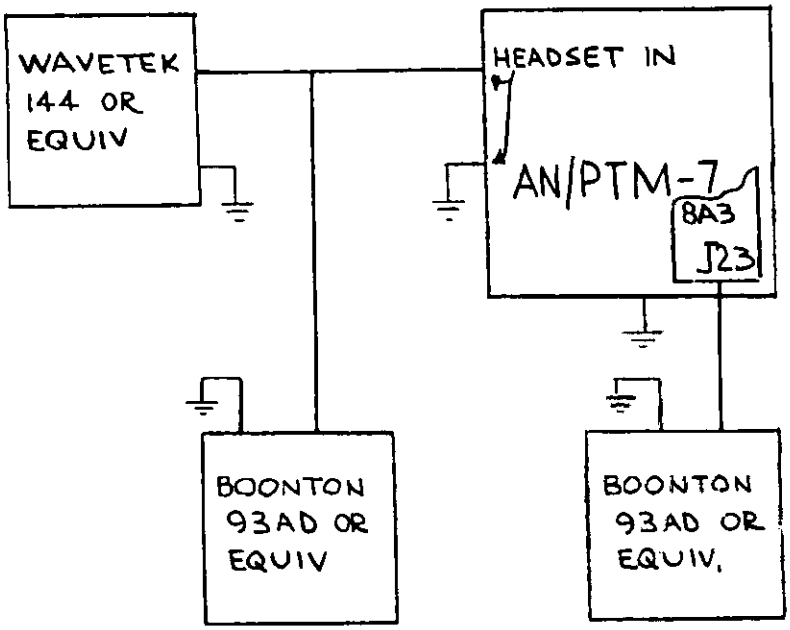
FIGURE 7E



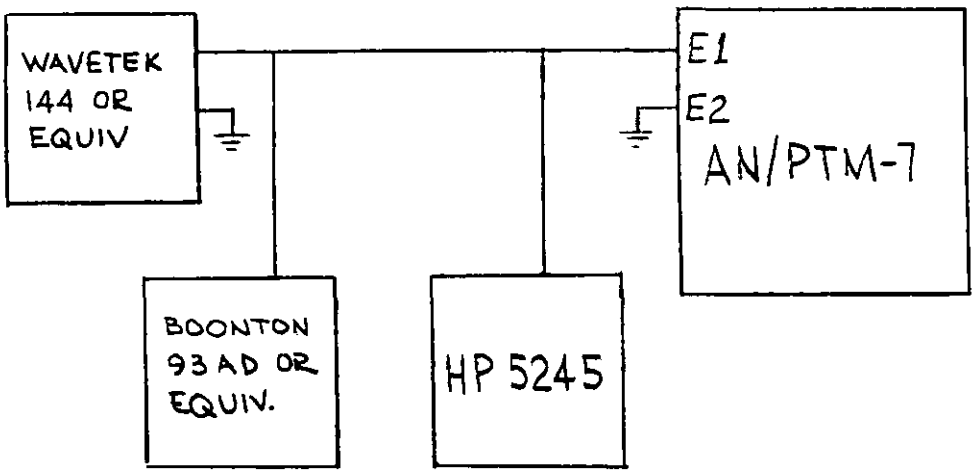
RECEIVER AMPLIFIER TEST

FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

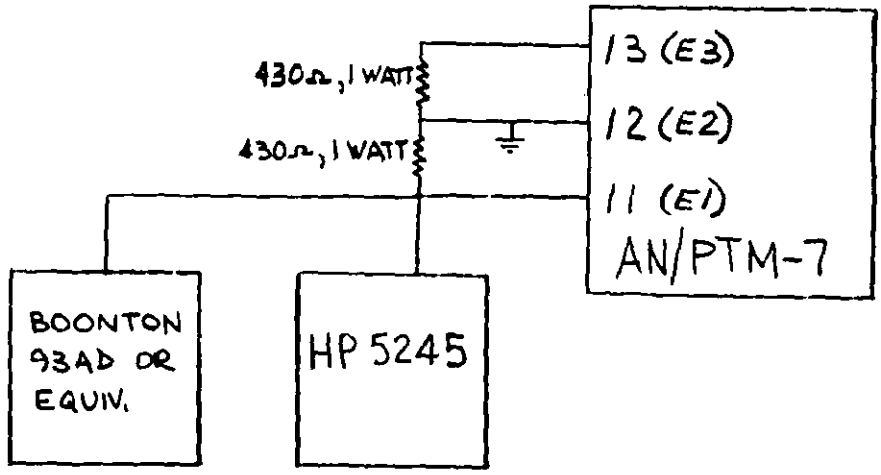


TRANSMIT AMPLIFIER TEST
FIGURE 9



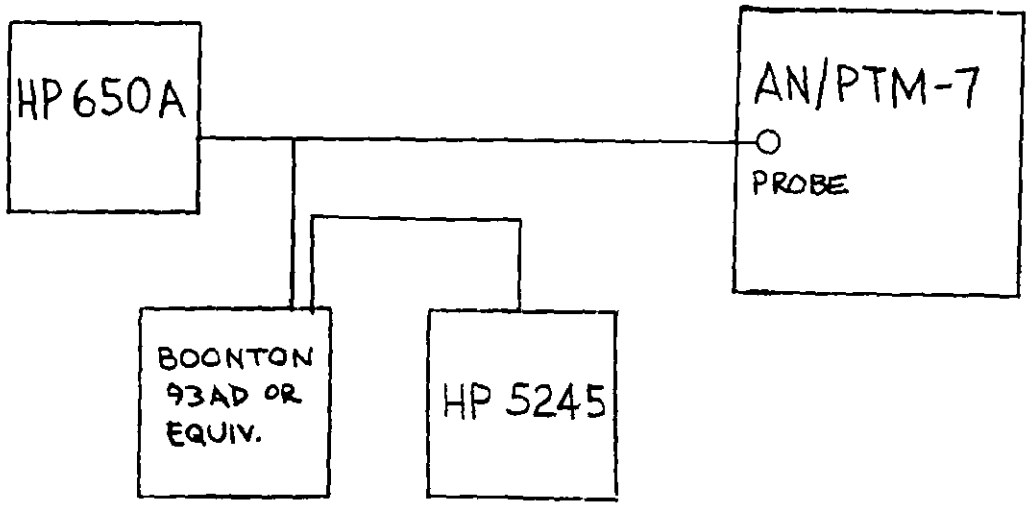
RINGING DETECTOR TEST
FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



SIGNAL AMP & OSCILLATOR TEST (PART I)

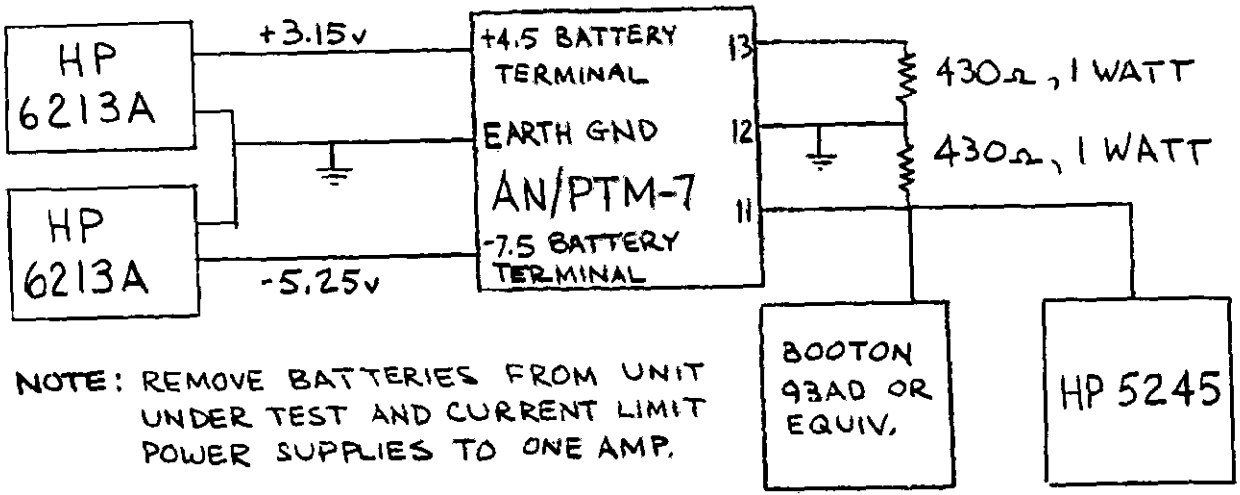
FIGURE 11



PCM DETECTOR TEST

FIGURE 12

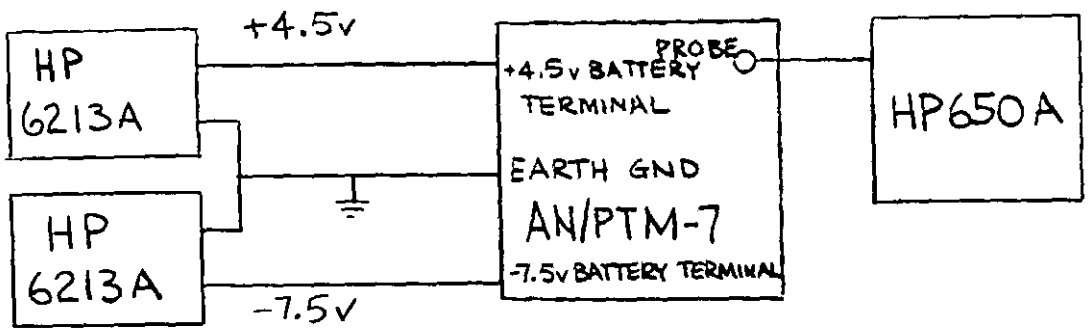
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



NOTE: REMOVE BATTERIES FROM UNIT UNDER TEST AND CURRENT LIMIT POWER SUPPLIES TO ONE AMP.

SIGNAL AMP & OSCILLATOR TEST (PART II)

FIGURE 13

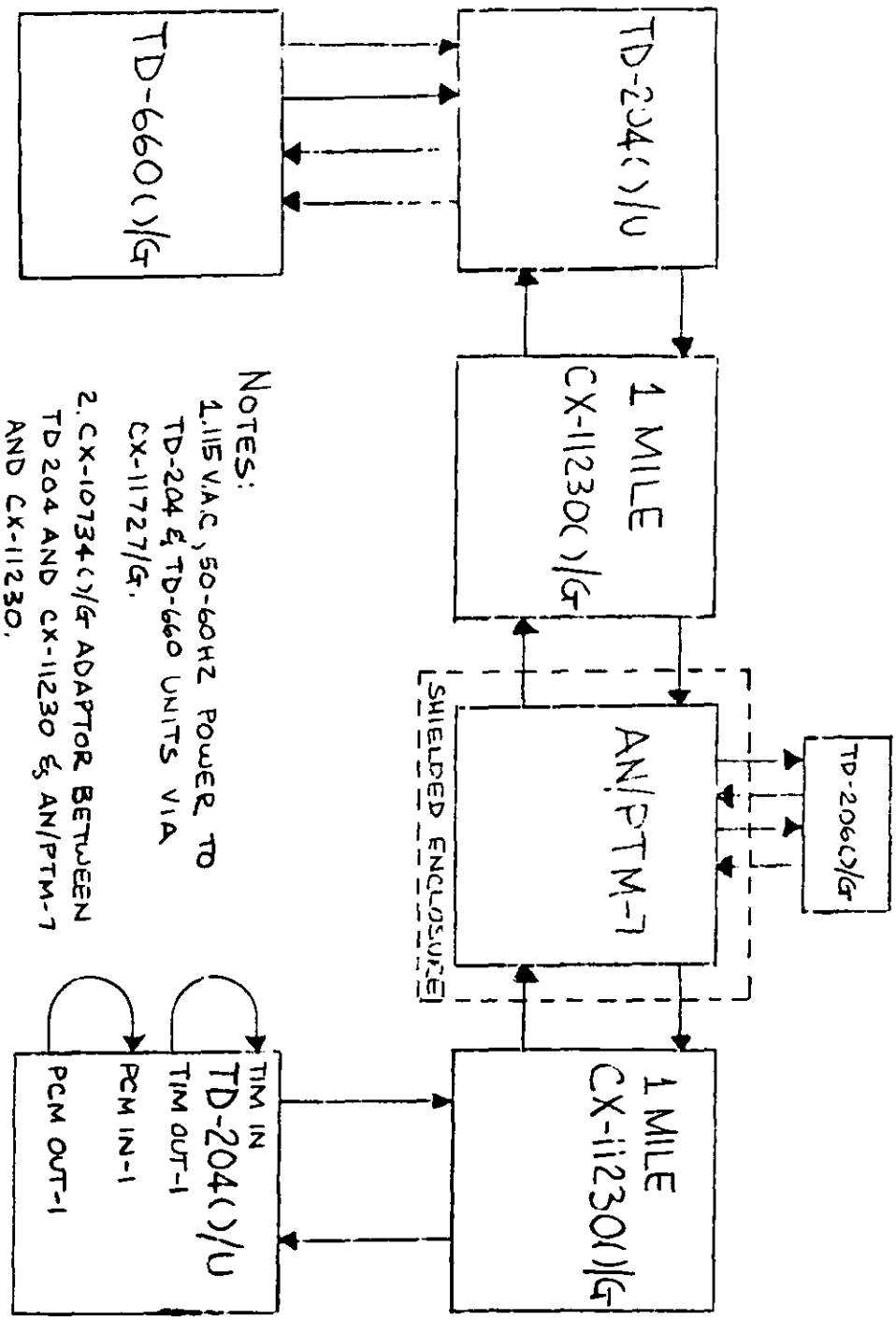


NOTE: REMOVE BATTERIES FROM UNIT UNDER TEST

ELECTRICAL PRECONDITIONING

FIGURE 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



ELECTROMAGNETIC COMPATIBILITY (EMI) TEST

FIGURE 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-RD-TS
Fort Monmouth, New Jersey 07703

To detach this form, cut along this line

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-55603A(EL) TEST SET, TELEPHONE AN/PTM-7( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.